PV-audit (Physical_Validator)

A Framework for Quantifying AI Robustness Under Physical Constraints

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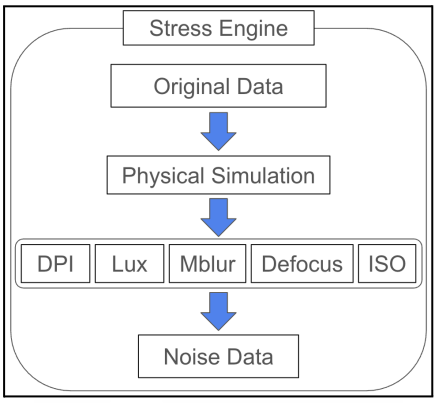
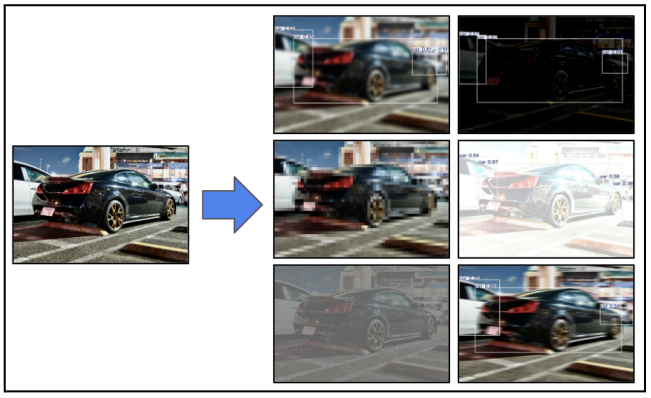
Abstract

Industrial AI deployment often faces a gap between performance on static benchmarks and actual behavior in production environments. To address this, we introduce the Production Validation (PV) framework, which treats AI models as black-box systems exposed to systematic physical stress testing. Using Confidence and IoU as key metrics, the framework evaluates model robustness against physical variables with stress engine[1] such as[2]:

- **Lux: Illumination variations**
- **DPI: Sensor resolution limits**
- **Motion Blur & Defocus: Optical and kinetic disturbances**
- **ISO: Sensor-level noise intensity**



[1]System Flow for Stress Engine

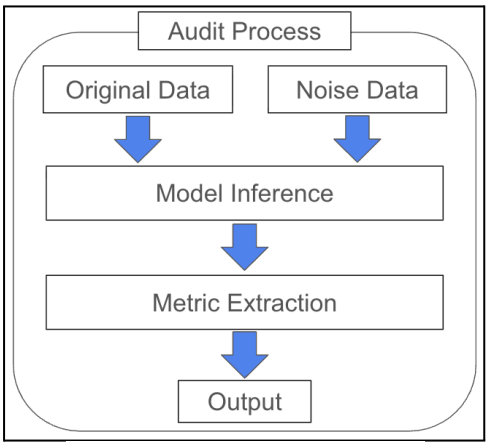


[2] Adding noise to the original image

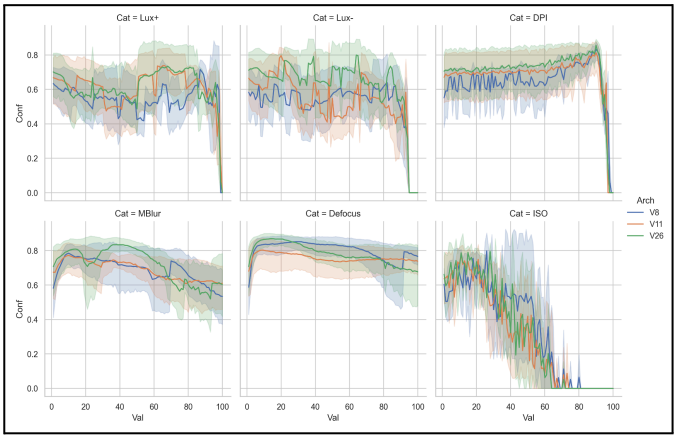
This enables the definition of objective operational boundaries (Red Lines) that directly inform hardware specification and quality assurance.

Methodology: Environmental Stress Testing

A stress engine systematically applies environmental stressors to datasets, mapping performance degradation curves across different model architectures[3][4]. This approach identifies the conditions under which models are likely to fail, allowing for early system-level mitigation such as automated error reporting or handover to human inspection.



[3] System Flow for Audit Process

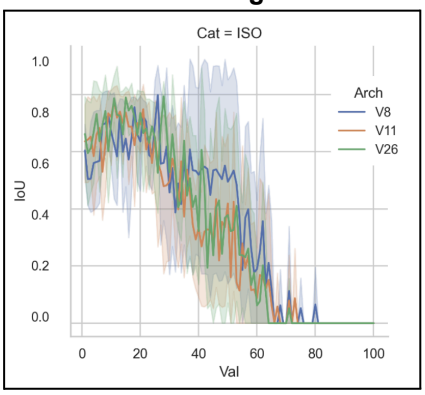


[4] Result from each model

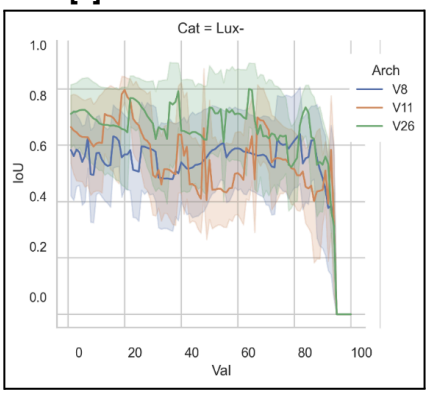
Key Observations (with yolov8 yolo11 yolo26)

1. **Performance Degradation Patterns**

- **Models exhibit diverse sensitivity profiles to environmental stressors. ISO noise is often a critical system-level failure trigger[5], and sensor physical limits can dominate architectural advantages under low-light conditions[6].**



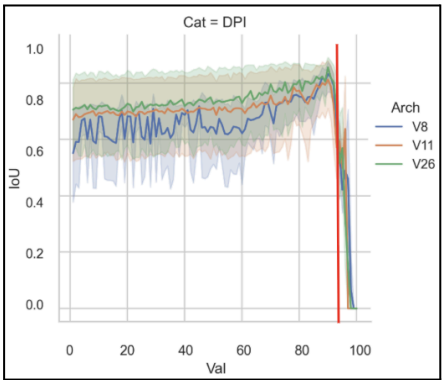
[5] each model's IoU in noise stress



[6] each model's IoU in low-light stress

1. **Precision and Stability**

- **Some architectures maintain high nominal accuracy but demonstrate varying stability thresholds under extreme physical conditions. If a model still outputs 90% confidence when DPI is reduced to the point of being barely human-recognizable[7][8], it indicates potential overfitting.**



[7] each model's confidence in low-DPI stress

[8] Noise data with low-DPI stress

1. **Boundary-Based Validation**

- **Transitioning from accuracy-focused evaluation to boundary-focused validation allows us to define an Operational Envelope, providing a deterministic basis for preventive maintenance and hardware selection. PV also enables quantitative assessment of the benefits from camera or sensor upgrades.**

Conclusion

The PV framework can shift industrial AI integration from purely statistical performance to quantifiable operational reliability. By systematically identifying failure thresholds, organizations can ensure scalable and repeatable quality assurance for AI systems in complex production environments.

Note: These observations and methods are derived from real-world experience in automotive OEM production lines, where environmental variability poses significant challenges to AI stability and reliability.